TEXORA · LECTURE SHEET

DigiEye System for Color Measurement

A calibrated camera in a controlled daylight cabinet that measures colour the way the eye sees it — pixel by pixel, across the whole sample, not just at one small aperture.

WP-4103 · COLOR PHYSICS · TOPIC 4 · SPECIALIZATION — WET PROCESSING

1. What DigiEye is — digital imaging colorimetry

A spectrophotometer measures one small area of a sample and reports spectral reflectance. The **DigiEye system** (developed by VeriVide) takes the complementary approach of **digital imaging colorimetry**: a calibrated digital camera photographs the sample inside a controlled-lighting cabinet, and software converts the image into CIE colour values for every pixel — a colour measurement *and* a true image of the sample at the same time. The system was designed for products whose colour cannot be reduced to one aperture reading: multicoloured prints, lace, knits, slub and mélange yarns, textured and pile fabrics, garments and trims (buttons, zips, labels) — anything where *where* the colour is matters as much as *what* it is.

2. System components

COMPONENT	FUNCTION
Lighting cabinet	Enclosed booth with diffuse **D65-simulating** illumination of controlled intensity and geometry, neutral interior — the standard viewing and imaging environment
Calibrated digital camera	Fixed-position camera with fixed exposure and white balance, imaging the sample under the cabinet light
Calibration / characterisation targets	Reference charts of known colour values used to build and verify the camera’s colour transform
Software	Converts image RGB to XYZ and L*a*b*, defines measurement areas, compares with the digital standard, reports ΔE and displays calibrated images for remote visual assessment
Calibrated monitor (optional)	Displays the captured colour accurately so a buyer in another country can visually approve the same image the mill measured

3. Characterisation and calibration — making a camera into an instrument

A consumer camera's RGB numbers are device-dependent: the same fabric gives different RGB values on different cameras. **Characterisation** converts the DigiEye camera into a measuring device:

1. The camera images a **reference chart** containing patches whose true CIE XYZ / L*a*b* values are known from spectrophotometric measurement.
2. Software fits a mathematical transform (typically a polynomial regression / matrix model) mapping camera RGB responses to the known XYZ values across the chart's colour range.
3. The transform is verified on independent check patches; residual errors must fall within the system's stated tolerance before measurement begins.
4. **Routine calibration** — imaging the calibration target at the start of work and at set intervals — corrects drift from lamp ageing and camera response, exactly as a spectrophotometer is calibrated on its white tile and black trap.

Because the whole chain (cabinet light, camera, transform) is characterised together, the captured image is itself colour data: every pixel carries calibrated XYZ and L*a*b* values under the cabinet's D65 condition.

4. From RGB to a colour report — the conversion chain

Sample under D65 cabinet light → camera RGB (per pixel) → characterisation transform → CIE XYZ → L*a*b* / L*C*h → ΔE versus digital standard

The XYZ-to-L*a*b* step is the standard CIE conversion of Topic 2 (normalising by the D65 white point), so DigiEye results are reported in the same L*a*b*, ΔE and tolerance language as spectrophotometer results and can sit in the same specification — typically D65/10°. The operator selects measurement regions on the image (a whole panel, one stripe of a print, a single button) and the software averages the pixels in each region, excluding shadows, creases and specular highlights where instructed.

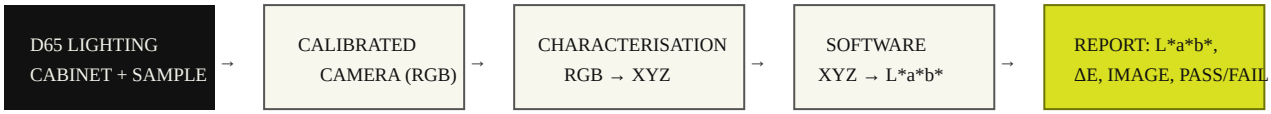


Figure 1 — DigiEye workflow: controlled D65 cabinet, calibrated camera, characterisation transform, CIE conversion and a combined numeric + image report.

5. DigiEye versus spectrophotometer

FEATURE	DIGIEYE (IMAGING)	SPECTROPHOTOMETER
Measurement basis	Calibrated camera image; RGB → XYZ → L*a*b*	Spectral reflectance curve R(λ), typically 10 nm steps
Area measured	Whole sample or any selected region, pixel by pixel	One fixed aperture (e.g. 5–30 mm), one spot per reading
Multicolour / patterned samples	Excellent — each colour area measured separately from one image	Difficult — aperture averages colours or must be repositioned colour by colour
Texture / appearance record	Yes — calibrated image retained with the numbers	No image; numbers only
Spectral data / match prediction	No spectrum — cannot drive Kubelka–Munk recipe prediction or compute metamerism across illuminants from first principles	Full spectrum — supports computer colour matching and multi-illuminant calculation
Small / curved / irregular items	Good — buttons, laces, garments imaged as they are	Poor — needs a flat area covering the aperture
Remote approval	Strong — buyer views the same calibrated image	Numbers can be sent, but appearance cannot be *seen*
Typical role	Shade approval, appearance and pattern assessment, virtual sampling	Recipe formulation, laboratory dyeing control, primary pass/fail reference

The instruments are complementary, not rivals: the spectrophotometer remains the reference for dye recipe work and spectral specification, while DigiEye answers the questions an aperture cannot — pattern, placement, texture and overall appearance.

6. Whole-sample imaging — texture and pattern assessment

Because measurement is per-pixel, DigiEye does more than average a shade:

- **Colour separation of prints:** each printed colour is selected and measured individually from one image of the pattern.
- **Uniformity mapping:** pixel statistics reveal unlevelness, shading side-to-side or end-to-end (listing), and barré — as a colour-difference map, not one mean value.
- **Texture and surface:** the calibrated image documents pile, slub, mélange distribution and surface effects alongside colour, so approvals cover appearance, not only coordinates.
- **Multi-component products:** shell fabric, lining, thread, buttons and zip are measured in their real positions in a single garment image — the basis of whole-garment shade approval.

7. Applications in production

- **Shade approval without couriers:** the mill images the bulk sample; the buyer receives L*a*b*/ΔE data *with the calibrated image* and approves or rejects digitally — days saved per submission round and fewer physical standards fading in transit.
- **Virtual sampling:** colourways are simulated and assessed on screen before physical samples are dyed, cutting sample yardage, chemicals and time in product development.
- **Supplier consistency:** the same digital standard and imaging condition are applied across factories and countries, removing cabinet-to-cabinet and observer-to-observer variation (Topic 3).
- **Records and disputes:** the time-stamped image plus numbers form an auditable approval record.

8. Limitations

- **No spectral data:** without R(λ), metamerism can only be checked by re-imaging under another light source, not calculated; recipe prediction (Kubelka–Munk, Topic 2) is impossible from image data alone.
- **Illuminant fixed by the cabinet:** imaging is performed under the cabinet's D65 simulation; accuracy depends absolutely on its quality and maintenance.
- **Fluorescence:** optically brightened whites respond to the UV content of the source; camera systems handle fluorescence less rigorously than UV-calibrated spectrophotometers.
- **Geometry and surface:** gloss, deep texture and three-dimensional items cast shadows and highlights; careful sample presentation and region selection are essential, and results remain geometry-dependent.

- **Calibration discipline:** the system is only as accurate as its characterisation and routine calibration; skipped calibration silently invalidates the data — the same discipline as any instrument (Topic 2).
- **Monitor dependence:** remote *visual* approval is valid only on a calibrated display; the numeric report does not have this dependency.

9. Glossary — key terms

- **Digital imaging colorimetry:** colour measurement from a calibrated camera image rather than a point spectrometer reading.
- **Characterisation:** building the mathematical transform from a camera's RGB responses to CIE XYZ using patches of known colour.
- **Calibration (routine):** re-imaging a reference target to correct instrument drift before measurement.
- **Digital standard:** a shade held as reference data (L*a*b* values, and in imaging systems a reference image) instead of only a physical swatch.
- **Virtual sampling:** creating and approving colourways digitally before making physical samples.

10. Key points

- DigiEye = controlled D65 cabinet + characterised camera + CIE conversion: every pixel becomes an L*a*b* measurement.
- Its strengths are whole-sample, multi-colour, texture and garment assessment plus image-based remote approval.
- Its hard limit is the absence of spectral data — no recipe prediction and no calculated metamerism; the spectrophotometer stays the laboratory reference.
- Accuracy rests on characterisation, routine calibration and disciplined sample presentation.

11. Summary

DigiEye extends colour measurement from a point to a picture. A calibrated imaging chain inside a D65 cabinet turns photographs into traceable CIE data, enabling per-colour measurement of prints, uniformity mapping, whole-garment approval and virtual sampling across a global supply chain. Used beside — not instead of — the spectrophotometer, it closes the gap between what an instrument measures and what a customer actually sees.

12. Practice

1. A buyer must approve a five-colour floral print and a matching solid lining. Explain why DigiEye is preferred for the print and which instrument should control the lining's dye recipe.
2. List, in order, the steps that convert a camera image in the cabinet into an L*a*b* report.
3. A DigiEye result is suspected of drifting after a lamp change in the cabinet. What two procedures restore confidence, and why?
4. State two DigiEye limitations that follow directly from its lack of spectral data.
5. Why can a buyer's on-screen visual approval be invalid even when the numeric ΔE report is sound?

Answer key

1. The print's colours are small, separate areas: one DigiEye image lets each colour be selected and measured individually, whereas a spectrophotometer aperture would average colours or need laborious repositioning. The lining's recipe should be controlled by spectrophotometer, because recipe prediction and correction need spectral R(λ) and Kubelka–Munk.
2. Image sample under cabinet D65 → capture camera RGB per pixel → apply characterisation transform RGB → XYZ → convert XYZ → L*a*b* (D65 white point) → compare with the digital standard and report ΔE and the calibrated image.
3. Routine calibration (image the calibration target to correct the chain's drift) and verification of the characterisation on independent check patches — a new lamp changes the SPD and camera response, so the RGB → XYZ relationship must be re-corrected and proved before measuring.
4. (Any two:) metamerism cannot be calculated for other illuminants — samples must be re-imaged under each light; computer colour matching / recipe prediction is impossible without R(λ); fluorescence/UV behaviour cannot be quantified spectrally.
5. Visual approval depends on the buyer's display; an uncalibrated monitor shifts the displayed colour even though the underlying numbers are correct. Calibrated monitors (or reliance on the numeric report) remove the risk.
